

Regional climate shocks, adaptation policy and distributional outcomes in an agent-based model

A regional DSK Macro-economic Agent-Based Model

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1. Introduction

2. Model Extension Summary

3. Regional Firms & Energy Plants

4. Regional Labour

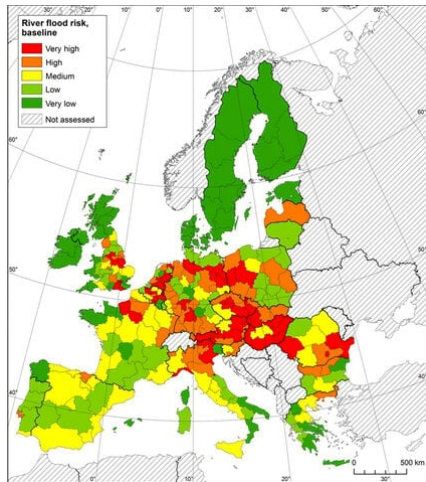
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Climate Damages Are Uneven *within* countries



Source: (Lung et al., 2012)

- ▶ Extreme weather events (EWEs) are increasing in frequency, intensity and economic cost (Intergovernmental Panel On Climate Change (Ipcc), 2023; Coronese et al., 2019)
- ▶ EWE exposure is geographically concentrated — also within countries (Lung et al., 2012; Coscarelli et al., 2021; Arrighi and Domeneghetti, 2023)
- ▶ Regions also differ in *recovery capacity*: (Noy, 2009)
- ▶ Repeated shocks to lagging regions can trigger structural divergence that is not captured in aggregate statistics (Strobl, 2011)

Aggregate Stats Mask Regional heterogeneity

- ▶ National GDP growth and recovery post EWE can look healthy while specific regions experience persistent decline (Strobl, 2011)
- ▶ Post-disaster GDP recovery can actively **mask** worsening social vulnerability, inequality and human development in affected regions (Clò et al., 2025)
- ▶ Regional divergence becomes self-reinforcing

Research Question

How do regionally heterogeneous extreme weather events shape macroeconomic dynamics and climate policy outcomes through opinion dynamics?

1. How does heterogeneous exposure to extreme weather events affect regional inequalities and macroeconomic outcomes?
▶ Phase 1
2. How do localized experiences of extreme weather events influence regional and national climate policy preferences, including both the strengthening and erosion of support for mitigation and adaptation policies?
▶ Phase 2

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Phase 1: Regionalised Agents and Shocks

- ▶ **Objective:** Explore how regional EWE exposures affect national and regional macro-economic outcomes.
- ▶ **Modelling:**
 - Regionalised agents: firms, labour (households), and energy plants
 - Capital Market, Goods Market and Energy Market remains at national level (unchanged). However, there is a flag to introduce home bias for locally produced machines and consumption goods.
 - Regional governments with budgets and policies.
- ▶ **Hypothesis:** Regional EWE damages create persistent economic path dependencies.
- ▶ **Policies:**
 - Baseline: no active regional governments.
 - Adaptation: regions offer relief subsidies post EWE or invest in protective adaptation for damage reduction.
 - Policy mixes: test interactions of national mitigation and regional adaptation.

Phase 2: Regional Shocks & OD

These are planned extensions which are not yet implemented

- ▶ **Objective:** Endogenize support for climate policies via opinion dynamics.
- ▶ **Modelling:**
 - Drivers: EWE exposure, attribution, beliefs, employment, (dis)information.
 - Outcomes: Differentiated support for mitigation (national) vs. adaptation (regional) policies.
- ▶ **Hypotheses:**
 - EWE exposure increases adaptation support; mitigation support is conditional
 - Repeated EWE can either exacerbate or alleviate political polarization and climate fatalism among residents.
- ▶ **Policies:**
 - Baseline: Households form opinions on national carbon policy.
 - Adaptation scenario: Opinions include regional adaptation support.
 - Election scenario: Opinion shifts only affect policies during election cycles.

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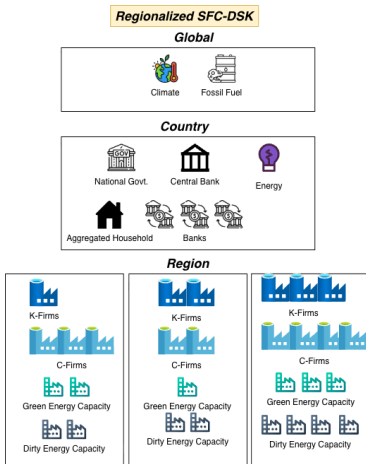
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Introducing Regional Layer

1. Region is conceptualized as an aggregator/accounting layer in the existing model. It is not an agent and does not influence model behaviour.
2. K-Firms, C-Firms and Green and Brown energy plants are tagged to a region based on proportional distribution specified in the parametrisation.
3. Energy capacity, production, emissions, etc. are computed at regional level as derivatives based on regional capacity share.
4. Other variables and behavioural aspects are left untouched.
5. Upon exit and entry, firms inherit the regional tag associated with the slot they occupy. *Proportion of K & C-firms does not evolve endogenously over the time.*
6. By default, both consumption and capital goods market remains at national level. We introduced a regional bias flag where consumption and capital goods produced in other regions are perceived as more expensive.

Regional Price Bias

► flag regional bias

0. buyers (households & C-Firms) perceive posted prices (baseline)
1. non-regional suppliers posted prices receive a *perceived* cost penalty; actual payments and stock-flow accounting are unchanged.

► Perceived price wedge for buyer i and supplier j :

$$\tilde{p}_{ij,t} = p_{j,t} \left[1 + \underbrace{\tau_R}_{\text{Additional Perceived Costs}} \cdot \mathbb{I}(r_i \neq r_j) \right], \quad \tilde{p}_{ij,t} = \begin{cases} p_{j,t}, & r_i = r_j, \\ (1 + \tau_R) p_{j,t}, & r_i \neq r_j \end{cases}$$

- This additional perceived cost τ_R (*parameter*) captures regional frictions (i.e. transport costs, local networks, and procurement familiarity)
- Regional home-bias affects *supplier choice only*. Actual transactions still occur at posted prices p_2 and p_1 , preserving SFC accounting.

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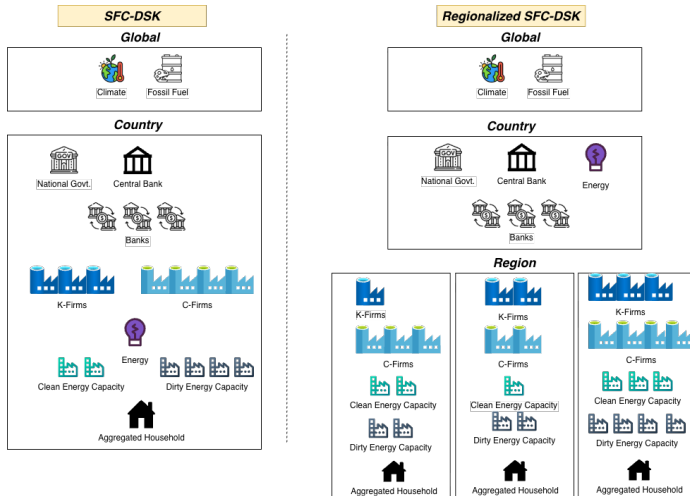
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Regional Labour / Households



Regionalising Households/Labour

► Regional Labour demand

$$\underbrace{LD_{r,t}}_{\text{Regional labour demand}} = \underbrace{\sum_{i \in \mathcal{K}_r} \ell_{1,i,t}^d}_{\text{K-firms labour demand}} + \underbrace{\sum_{j \in \mathcal{C}_r} \ell_{2,j,t}^d}_{\text{C-firms labour demand}} + \underbrace{LD_{r,t}^{rd}}_{\text{R\&D labour demand}} + \underbrace{LD_{r,t}^{en}}_{\text{Energy sector labour demand}}$$

► Regional labour supply is initialised based on labour supply share σ_r

$$\underbrace{LS_{r,t}}_{\text{Regional labour supply}} = \underbrace{LS_t}_{\text{National labour supply pool}} \cdot \underbrace{\sigma_r}_{\text{Parameter: region's labour-supply share}}, \quad \sum_{r=1}^{N_r} \sigma_r = 1$$

► Unemployment rate per region

$$u_{r,t} = \frac{LS_{r,t} - L_{r,t}}{LS_{r,t}}, \quad L_{r,t} = \min(LD_{r,t}, LS_{r,t})$$

Regional Labour Market

- If regional production labour demand exceeds available labour, ϕ_r scales firm-level labour use and output proportionally to the regional labour constraint (consistent with DSK implementation).

$$\text{if } \underbrace{\sum_{j \in C_r} \ell_{2,j,t}^d + \sum_{i \in K_r} \ell_{1,i,t}^d}_{\text{Production labour demand for region } r} > \underbrace{LSe_{r,t}}_{\text{Labour available for production in region } r} \Rightarrow \phi_r = \frac{LSe_{r,t}}{\underbrace{\sum_{j \in C_r} \ell_{2,j,t}^d + \sum_{i \in K_r} \ell_{1,i,t}^d}_{\text{Regional labour-rationing factor}}},$$

$$\underbrace{\ell_{\cdot,r,t}^d \leftarrow \phi_r \cdot \ell_{\cdot,r,t}^d}_{\text{Labour demand and therefore output scaled proportionately}}$$

$$\underbrace{LSe_{r,t}}_{\text{Labour available for production}} = \underbrace{LS_{r,t}}_{\text{Regional labour supply}} - \underbrace{LD_{r,t}^{rd}}_{\text{R\&D labour demand}} - \underbrace{LD_{r,t}^{en}}_{\text{Energy sector labour demand}}$$

National Wage in DSK Model

$$\underbrace{\dot{w}_t^{nat}}_{\text{National wage growth}} = \underbrace{\pi^*}_{\text{Inflation target}} + \underbrace{\psi_1(\pi_t - \pi^*)}_{\text{Inflation-gap effect}} + \underbrace{\psi_2 \dot{A}_t^m}_{\text{National productivity growth effect}} - \underbrace{\psi_3 \Delta u_t}_{\text{National unemployment effect}}$$

Aggregate wage update in DSK model (Reissl et al., 2025)

$$\underbrace{w_t}_{\text{National wage rate}} = \underbrace{w_{t-1}}_{\text{Lagged national wage rate}} \left(1 + \underbrace{\dot{w}_t^{nat}}_{\text{National wage growth}} \right)$$

Regional Wage in R-DSK

$$\underbrace{\dot{w}_{r,t}^{reg}}_{\text{Regional wage growth}} = \underbrace{\pi^*}_{\text{Inflation target}} + \underbrace{\psi_1(\pi_t - \pi^*)}_{\text{Inflation-gap effect}} + \underbrace{\psi_2 \dot{A}_{r,t}^m}_{\text{Regional productivity growth effect}} - \underbrace{\psi_3 \Delta u_{r,t}}_{\text{Regional unemployment change effect}}$$

$$\underbrace{A_{r,t}^m}_{\text{Regional mean productivity}} = \frac{\sum_{i \in \mathcal{K}_r} A_{i,t}^K + \sum_{j \in \mathcal{C}_r} A_{j,t}^C}{N_r^K + N_r^C}$$

$$\underbrace{\dot{A}_{r,t}^m}_{\text{Regional productivity growth}} = \underbrace{\kappa}_{\text{Smoothing parameter}} \dot{A}_{r,t-1}^m + (1 - \kappa) \frac{A_{r,t}^m - A_{r,t-1}^m}{A_{r,t-1}^m}$$

$$\underbrace{\Delta u_{r,t}}_{\text{Regional unemployment change}} = \frac{u_{r,t} - u_{r,t-1}}{u_{r,t-1}}$$

Regional Wage in R-DSK

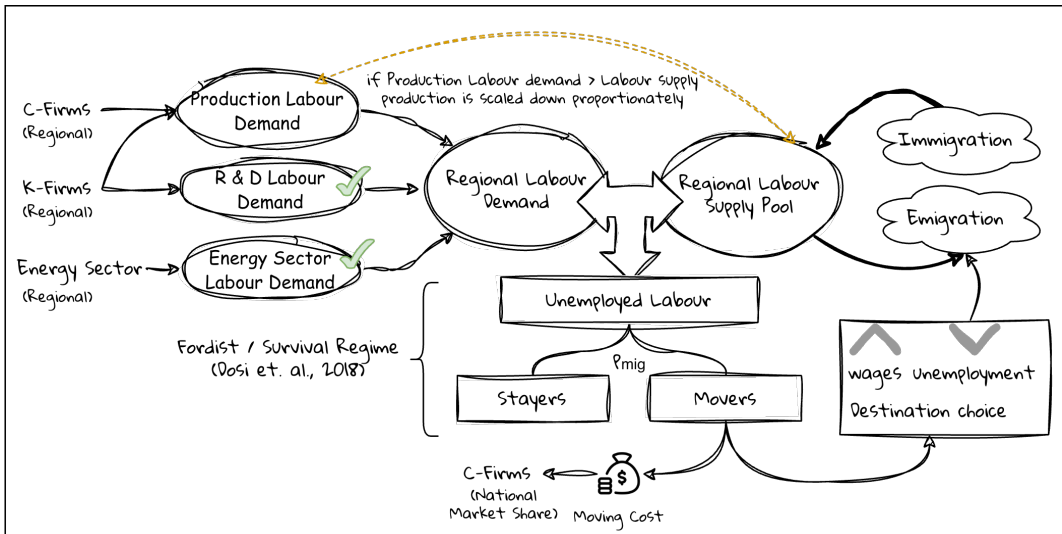
$$\underbrace{\dot{w}_{r,t}}_{\text{Actual regional wage growth}} = \underbrace{\chi \dot{w}_t^{nat}}_{\text{National wage-setting component}} + \underbrace{(1 - \chi) \dot{w}_{r,t}^{reg}}_{\text{Regional wage-setting component}}, \quad |\dot{w}_{r,t}| \leq \bar{\dot{w}}$$

$\chi = 1$: Fully national wage growth

$\chi = 0$: Fully regional wage growth

$$\underbrace{w_{r,t}}_{\text{Regional wage rate}} = \underbrace{w_{r,t-1}}_{\text{Lagged regional wage rate}} \left(1 + \underbrace{\dot{w}_{r,t}}_{\text{Actual regional wage growth}} \right)$$

Regional Labour Migration



Regional Attractiveness ($V_{r,t}$)

- Unemployed households compare regions according to regional real wages and unemployment conditions (Morten and Oliveira, 2016; Roback, 1982).

$$\omega_{r,t} = \frac{w_{r,t}}{p_t}, \quad \tilde{u}_{r,t} = \max\{u_{r,t}, u_{\min}\}.$$

$$V_{r,t} = \underbrace{\beta_w \ln(\omega_{r,t})}_{\substack{\text{higher real wages} \\ \text{raise attractiveness}}} - \underbrace{\beta_u \ln(\tilde{u}_{r,t})}_{\substack{\text{higher unemployment} \\ \text{reduces attractiveness}}}, \quad \beta_w, \beta_u > 0.$$

- $u_{\min}$ prevents the logarithm from becoming undefined when regional unemployment approaches zero.

Heterogeneous Willingness to Move

- ▶ Only unemployed households form the potential migration pool.
- ▶ Households differ in their willingness to relocate through a heterogeneous migration threshold:

$$\tau_h \sim \Gamma \left(k_\tau, \frac{\bar{\tau}}{k_\tau} \right).$$

- ▶ For a household currently in region o , destination d enters its choice set only if

$$V_{d,t} - V_{o,t} > \tau_h.$$

- ▶ Higher τ_h represents stronger attachment or non-monetary migration frictions.
- ▶ The threshold determines which destinations are considered; it does not determine the final destination itself.

Destination Choice and Intended Migration

- Households choose probabilistically between staying and the admissible destinations.

$$\pi_{o,d,t}^h = \frac{\exp(V_{d,t})}{\exp(V_{o,t}) + \sum_{k \in \mathcal{D}_{o,h,t}} \exp(V_{k,t})}, \quad d \in \mathcal{D}_{o,h,t},$$

$$\pi_{o,o,t}^h = \frac{\exp(V_{o,t})}{\exp(V_{o,t}) + \sum_{k \in \mathcal{D}_{o,h,t}} \exp(V_{k,t})}.$$

- Destination probabilities are averaged across the heterogeneous migration-threshold distribution.
- Intended bilateral migration is then

$$M_{o,d,t}^{int} = UN_{o,t} \pi_{o,d,t}, \quad d \neq o.$$

Monetary Moving Cost

- Migration entails a fixed monetary cost proportional to expected annual labour income in the origin region (Morten and Oliveira, 2016; Fischer, 2019).
- Moving costs are independent of destination distance in the current implementation.

$$\underbrace{mc_{o,d,t}}_{\text{cost per migrant}} = \underbrace{\mu_F \mathbb{I}(o \neq d)}_{\text{fixed moving cost parameter}} \times \underbrace{4w_{o,t}(1 - u_{o,t})}_{\text{annual expected origin income}}.$$

$$ME_{o,t}^{int} = \sum_{d \neq o} mc_{o,d,t} M_{o,d,t}^{int}.$$

- Hence moving costs affect not only migration incentives, but whether intended migration is financially feasible.

Liquidity-Constrained Migration

- ▶ A desire to migrate does not guarantee that the move is financially feasible.
- ▶ Available migration liquidity is approximated by the share of regional household deposits associated with unemployed households:

$$D_{o,t}^{h,U} = u_{o,t} D_{o,t}^{h,\text{pre}}.$$

$$\lambda_{o,t}^{\text{liq}} = \min \left\{ 1, \frac{D_{o,t}^{h,U}}{ME_{o,t}^{\text{int}}} \right\}.$$

$$M_{o,d,t} = \lambda_{o,t}^{\text{liq}} M_{o,d,t}^{\text{int}}.$$

- ▶ If sufficient liquidity is available, $\lambda_{o,t}^{\text{liq}} = 1$ and all intended migration occurs.
- ▶ Otherwise, bilateral outflows are proportionally rationed.

Migration and SFC Closure

- Actual migration redistributes the national labour supply across regions:

$$\tilde{L}S_{r,t+1} = LS_{r,t} - \underbrace{\sum_{d \neq r} M_{r,d,t}}_{M_{r,t}^{out}} + \underbrace{\sum_{o \neq r} M_{o,r,t}}_{M_{r,t}^{in}}.$$

$$s_{r,t+1}^{LS} = \frac{\tilde{L}S_{r,t+1}}{\sum_q \tilde{L}S_{q,t+1}}.$$

- Migration decisions made in period t affect regional labour supply only from $t + 1$ onward.
- Moving expenditure is an explicit monetary flow: households pay the cost and C-firms receive migration-service revenue (distributed based on their national market share).
- Hence migration redistributes financial stocks without creating or destroying financial assets.

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Climate Shocks in DSK

Climate shocks are modelled as stochastic shocks drawn from temperature-dependent Beta distributions in Reissl et al. (2025).

For each shock channel $s \in \{1, \dots, S\}$ and period t :

$$\varepsilon_{s,t} \sim \text{Beta}(a_{s,t}, b_{s,t}), \quad \varepsilon_{s,t} \in (0, 1)$$

The shape parameters evolve with global temperature T_t :

$$a_{s,t} = a_{0,s} \left[1 + \ln \left(\frac{T_{t-1}}{T_0} \right) \right]^{\theta_s^{(1)}}$$

$$b_{s,t} = b_{0,s} \left(\frac{T_0}{T_{t-1}} \right)^{\theta_s^{(2)}}$$

- ▶ $a_{0,s}, b_{0,s}$ are baseline shape parameters
 - $a_{s,t}$ governs mean/median of damages
 - $b_{s,t}$ governs right-tail thickness (extreme events)
- ▶ $\theta_s^{(1)}$ controls increase in typical damages
- ▶ $\theta_s^{(2)}$ controls amplification of extremes
- ▶ T_t is global mean temperature, T_0 is reference level
- ▶ Climate shocks affect real economic variables multiplicatively:

$$X_{i,t} = X_{i,t-1} (1 - \varepsilon_{s,t})$$

Parameters of β curves for drawing shocks

1. **Global temperature** (T_t) is common to all regions
2. **Shock exponent** $\theta_s^{(1)}$ **and baseline severity parameters** $a_{0,s}$ have regional values but are kept same common across regions as they reflect physical damage sensitivities (how temperature translates into damage)
3. **Baseline exposure** ($b_{0,r}$): Region-specific right-tail thickness capturing structural exposure to extreme events (e.g. coastal vs inland regions).
4. **Extreme-event amplification** ($\theta_r^{(2)}$): Region-specific scaling of tail risk as temperature rises, allowing extreme damages to accelerate faster in some regions.

Regionalisation of Climate Shock Channels

- ▶ In the regional extension, climate shocks are generated *at the regional level* and propagated to all firms located in that region.
- ▶ For each shock channel $s \in \{1, \dots, S\}$, region r , and period t :

$$\varepsilon_{s,r,t} \sim \text{Beta}(a_{s,r,t}, b_{s,r,t})$$

- ▶ The realised regional shock $\varepsilon_{s,r,t}$ is then broadcast to all firms (or specified channel) i in region r :

$$X_{i,t} = X_{i,t-1}(1 - \varepsilon_{s,r,t}), \quad i \in r$$

- ▶ This mechanism is identical to Reissl et al. (2025).

Regionalisation of climate shock channels

- ▶ At each period, we draw one climate shock per region and per channel, driven by a common global temperature path and applied uniformly to all firms in the region.
- ▶ Regions differ in how strongly specific climate shock types hit (damages) and scale with warming, but all regions share the same global climate forcing.

Climate Shock Channels in DSK

► Climate Shock Channels

1. Shocks to productivity affecting the characteristics of capital vintages `flag_prodshocks1`
2. Shocks to productivity (not affecting characteristics of capital vintages) `flag_prodshocks2`
3. Energy Capacity both green and brown `flag_encapshocks`
4. Shock to labour population `flag_popshocks`
5. Shock to aggregate consumption demand `flag_demandshocks`
6. Shock to C-Firms capital stock `flag_capshocks`
7. Output shocks to C & K Firms `flag_outputshocks`
8. Shock to C-Firms inventories `flag_inventshocks`
9. Shock to R & D of K Firms `flag_RDshocks`

► All firms in the region receive uniform shocks

- `flag_uniform_shock = 1`

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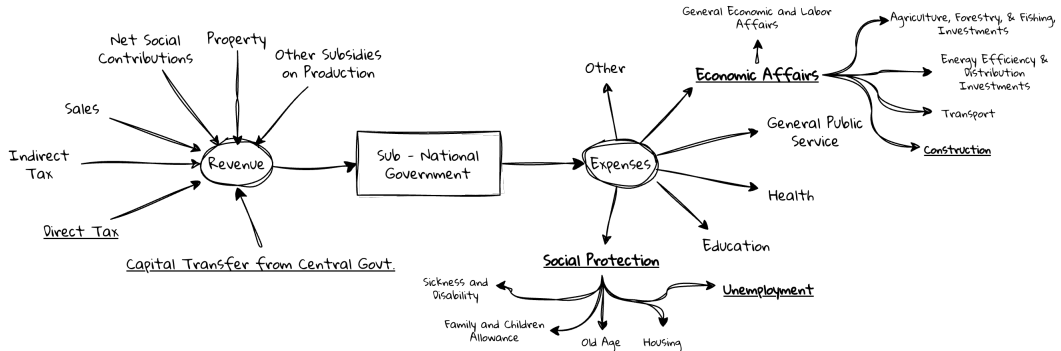
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Conceptualising Sub-National Government

- ▶ Each country creates its own arrangement for dividing responsibilities between regional and local entities which can be collectively called as sub-national governments (Halaskova et al., 2021)
- ▶ Expenses and revenues of sub-national government can be calibrated using subnational government finances from ECB Olsson and Catz (2023)



Regional Government Revenue Design

$$TaxShare_{r,t} = \tau_r^{share} \left(T_{r,t}^{Cfirms} + T_{r,t}^{Kfims} \right)$$

$$GRANTPOOL_t = \bar{\gamma} \cdot REVENUE_{CentralGovt,t}$$

$$GovtTransfer_{r,t}^{base} = \omega_r \cdot GRANTPOOL_t$$

$$GovtTransfer_{r,t}^{topup} = \max \left\{ 0, SocialProtection_{r,t} - \left(TaxShare_{r,t} + GovtTransfer_{r,t}^{base} \right) \right\}$$

$$REVENUE_{RegGovt,r,t} = GovtTransfer_{r,t}^{base} + GovtTransfer_{r,t}^{topup} + TaxShare_{r,t}$$

- ▶ $TaxShare_{r,t}$: Region's share of centrally collected taxes
- ▶ $GovtTransfer_{r,t}^{base}$: Standard intergovernmental transfer from grant pool
- ▶ $GovtTransfer_{r,t}^{topup}$: Additional transfer ensuring minimum welfare provision
- ▶ $REVENUE_{RegGovt,r,t}$: Total regional government revenue
- ▶ National government collects all taxes and redistributes to regions
- ▶ Grant pool bounded: $\bar{\gamma} \in [0, 1]$
- ▶ Allocation weights: $\sum_r \omega_r = 1$
- ▶ Welfare guarantee: $REV_{rg,r,t} \geq SP_{r,t}^{plan}$

Regional Government Expenditure Design

$$EXPENDITURE_{RegGovt,r,t} = SocialProtection_{r,t} + EconomicAffairs_{r,t}$$

Social Protection

$$SocialProtection_{r,t} = LS_{r,t} \cdot U_{r,t} \cdot w_{t-1} \cdot \omega_{u,r}$$

- ▶ $LS_{r,t}$: Regional labour supply pool
- ▶ $U_{r,t}$: Regional unemployment rate
- ▶ w_{t-1} : Wages
- ▶ $\omega_{u,r}$: Regional wage replacement rate

Economic Affairs

$$EconomicAffairs_{r,t} = \max \{0, REVENUE_{r,t} - SocialProtection_{r,t}\}$$

- ▶ Public investment expenditure for building public stock
- ▶ For each K-firm in region r :

$$EconomicAffairs_{Kfirm,r,t} = \frac{EconomicAffairs_{r,t}}{N_{r,t}^{Kfirm}}$$

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Recovery expenditure: mechanism

1. **Obligation is recognised.** A new recovery expense is booked:

$$I_{r,t}^{Rec} = \psi_r^A \cdot Y_{r,t-1} \cdot \max(0, \varepsilon_{r,t-1}),$$

2. **Per-firm allocation.** Equal split across affected firms.
3. **National-government top-up** If regional funds insufficient, national gov. tops up:

$$T_{r,t}^{topup} = \max\{ 0, I_{r,t}^{rec} - \max\{0, REV_{r,t}^{RG} - SP_{r,t}\} \}$$

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Protective adaptation (ex-ante)

Intuition

- ▶ Governments invest in adaptation before shocks hit
→ protection accumulates as a capital stock over time (Estrada et al., 2019; Bruin et al., 2007)
- ▶ Wealthier regions can sustain larger protection budgets (Saeed et al., 2023)
- ▶ Protection is probabilistic: small shocks are fully absorbed; large shocks still pass through but are dampened (Berkhout and Dow, 2023)

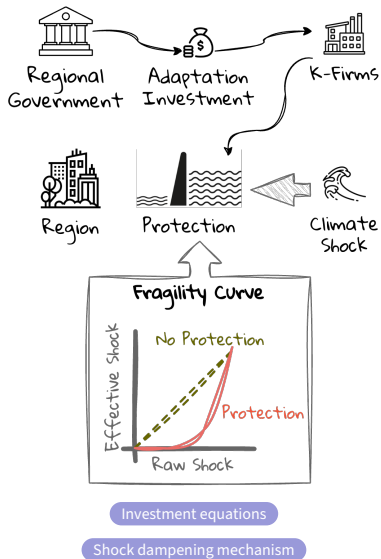
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Modelling

- ▶ Regional government of North invests $\iota^A = 2.5\%$ of their GDP every period = demand for local K-firms
- ▶ Shock dampening → **fragility curve** (Huizinga et al., 2017)



Shock absorption & dampening mechanism

Damage scales non-linearly with hazard intensity, with small events fully absorbed and large events passing through (Huizinga et al., 2017)

Effective shock (fragility curve)

$$\varepsilon_{r,t}^{\text{eff}} = \omega_{r,t} \cdot \varepsilon_{r,t}^{\text{raw}}, \quad \varepsilon_{r,t}^{\text{raw}} \sim \text{Beta}(a_{r,t}, b_{r,t})$$

$$\omega_{r,t} = \begin{cases} 0 & \varepsilon_{r,t}^{\text{raw}} \leq h_{r,t} \\ \left(\frac{\varepsilon_{r,t}^{\text{raw}} - h_{r,t}}{1 - h_{r,t}} \right) & \varepsilon_{r,t}^{\text{raw}} > h_{r,t} \end{cases}$$

Protection threshold

$$h_{r,t} = \bar{h}_r \left[1 - \exp \left(-\frac{K_{r,t-1}^A}{\gamma_{r,t}^n} \right) \right] \in [0, \bar{h}_r]$$

- If $K_r^A = 0 \Rightarrow h = 0$: no protection
- If K_r^A large $\Rightarrow h \rightarrow \bar{h}_r$: maximum absorption, ceiling $\bar{h}_r < 1$
- Uses lagged stock: investment only takes effect next period

Regional Ex-Ante / Planned Adaptation Investment

$$I_{r,t}^{Prot} = \iota_r^A \cdot Y_{r,t-1}$$

- ▶ ι_r^A : exogenous region-specific planned protection intensity (share of lagged real output); scaled so that larger/wealthier regions sustain proportionally larger protection budgets
- ▶ Implemented as a government order placed with K-firms in the region: the government credits K-firm deposit accounts by $I_{r,t}^{Prot} / N_{1,r,t}$ per firm, generating demand for physical protection capital (flood defences, drainage, resilient infrastructure) in the capital-goods sector

Protection (adaptation) capital accumulation

$$K_{r,t}^{Prot} = (1 - \delta_{Prot}) K_{r,t-1}^{Prot} \cdot I_{r,t}^{Prot}$$

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